DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 3 Winter 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTECPC302 - ELECTRONICS DEVICES & CIRCUITS

Time : 3 Hours] **[Total Marks : 60**

Instructions to the Students:

1.Each question carries 12 marks.
2.Question No. 1 will be compulsory and include objective-type questions.
3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4.Use of non-programmable scientific calculators is allowed.
5.Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question) 12

1 The JFET is a _______ controlled device.
A. Current
B. Voltage
C. Power
D. None

2 Pinch-off occurs when:
A. Drain current becomes zero
B. Channel is completely cut off
C. Drain current becomes constant
D. Gate draws current

3 In an n-channel E-MOSFET, the channel is formed by:
A. Doping
B. Inversion
C. Diffusion
D. Pinch off

4 The input stage of an op-amp is usually a:
A. Common emitter stage
B. Differential amplifier
C. Push-pull stage
D. Class C power stage

5 A voltage follower has a gain of:
A. Greater than 1
B. Less than 1
C. Exactly 1
D. Depends on resistors

6 An ideal integrator uses:
A. Capacitor at input and resistor in feedback
B. Resistor at input and capacitor in feedback
C. Inductor in input
D. Transformer in feedback

7 Voltage-series feedback results in:
A. High input impedance
B. Low input impedance
C. High output impedance
D. Decreased bandwidth

8 With negative feedback, bandwidth becomes:
A. Smaller
B. Larger
C. Same
D. Zero

9 Total gain of cascade amplifier is:
A. Sum of stage gains
B. Difference
C. Average of stage gains
D. Product of stage gains

10 A discrete transistor voltage regulator mainly uses:
A. Transistor + Zener diode
B. MOSFET + capacitor
C. Inductor + resistor
D. Relay + diode

11 Major disadvantage of a shunt regulator is:
A. Complex design
B. High efficiency
C. Low efficiency (current always flows)
D. No voltage regulation

12 Thermistors are based on:
A. Piezoelectric effect
B. Change of resistance with temperature
C. Magnetic induction
D. Optical absorption

Q2. Solve the following.

A) Draw and explain the circuit arrangement for VI characteristics of N-channel JFET and also draw the characteristic by showing the different regions. 6

B) A JFET with parameters I_{DSS}= 12 mA, $V_{(GS(off))}$= - 6 V, that provides V_{GS}= - 3 V. Determine the value of drain current I_D. 6

Q3. Solve the following.

A) Explain the principle of MOSFET in enhancement mode with neat sketches and also draw the output (drain) characteristics. 6

B) Derive the differential mode gain of a dual-input balanced-output differential amplifier. 6

Q4. Solve Any Two of the following.

A) Explain the inverting and non-inverting amplifier configurations with gain expressions. 6

B) Explain the working of an Op-amp integrator with proper equations. 6

C) Explain the concept of feedback in amplifiers and describe the four feedback topologies with diagrams. 6

Q5. Solve Any Two of the following.

A) Describe the working of an RC phase-shift oscillator and derive formula for frequency of oscillation. 6

B) Describe the construction and operation of a three-terminal 78xx series IC voltage regulator. 6

C) Explain the working of a shunt voltage regulator and list its advantages and disadvantages. 6

Q6. Solve Any Two of the following.

A) A discrete transistor voltage regulator uses a Zener diode of 8.2 V as reference.The transistor has V_{BE}= 0.7 V. If the load draws 100 mA, find the regulated output voltage and the power dissipated in the transistor when V_{in}= 15 V. 6

B) State and explain the essential requirements of a good sensor/transducer. 6

C) Explain the working principle of an LVDT (Linear Variable Differential Transformer) with neat diagram. 6

*** **End** ***